

Computational Fluid Dynamics - 7/7/2022

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Exercise 1. We want to simulate the flow of an incompressible fluid around a square obstacle of side $l = 0.2$, as represented in Figure 1, left. The kinematic viscosity of the fluid is $\nu = 0.002$. The problem is governed by the time-dependent incompressible Navier-Stokes equations. We consider the following boundary conditions:

- a uniform horizontal velocity profile is imposed on the inflow boundary Γ_4 : $\mathbf{u} = (1, 0)$;
- zero normal **stress** is imposed on the outflow boundary Γ_2 ;
- free-slip velocity conditions (equivalent to symmetry conditions) are imposed on the lower and upper boundaries Γ_1 and Γ_3 : $u_y = 0$, $\frac{\partial u_x}{\partial \mathbf{n}} = 0$;
- a no-slip condition is imposed on the square obstacle Γ_5 : $\mathbf{u} = (0, 0)$;

The problem is initialized considering, at time $t = 0$, the Stokes solution obtained prescribing the same boundary conditions as above.

The problem should be solved using the incremental Chorin-Temam projection method **in differential form**, using P_2/P_1 finite elements for velocity and pressure and advancing in time until the instant $T_f = 20$ with a time step $\Delta t = 0.1$.

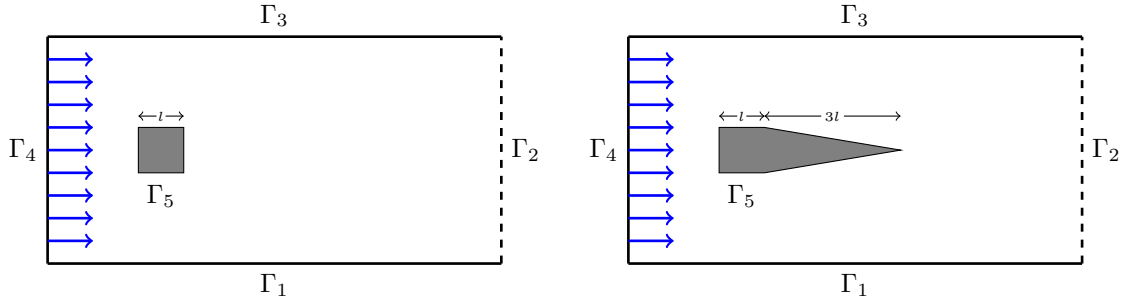


Figure 1: Original (left) and modified (right) computational domain

- a) Introduce the incremental Chorin-Temam projection method for the solution of the problem, discussing in detail the boundary conditions for each sub-problem.
- b) Write the weak formulation and the finite-element approximations of each sub-step of the projection method.
- c) Based on the template `exam.edp`, write a **FreeFEM++** script for the solution of the problem until the final time $T_f = 20$ and compute the time history of the lift and drag forces on the square obstacle.
- d) Adapt the **FreeFEM++** script to simulate the flow around the modified obstacle geometry displayed in Figure 1, right. Compute the time history of the lift and drag forces on the modified obstacle and discuss the differences with respect to the previous case.